Labasa Production System

What changed: revisions 1 to 7, commit `1e351c1`, 8 November 2026

#	Revision	Change
1	**Manage sizes**	New Ukuran master data page. New sizes appear on product forms automatically.
2	**Pick sizes per batch**	Tick the sizes to produce, each with its own quantity.
3	**Production log + Pause**	Pause records day, date and time in a chronological log.
4	**Wording**	Consistent, formal Indonesian across views, messages and validation.
5	**Machine ID**	Unique generated `kode_mesin` (MSN-001) replaces the old editable code.
6	**Buttons at the bottom**	Action buttons moved below the fields.
7	**Fuzzy stock**	Persediaan comes from product stock and is read-only.

How to read this report

- **Before** is the previous commit (`745f431`). **After** is the current code. Both ran side by side in Chrome.
- Red outlines were added to the screenshots to point out what changed. They are not part of the application.
- Screenshots use demo data from `DatabaseSeeder` on temporary databases. The batch was moved to the cutting stage directly in the temporary database so its stage form is visible.
- The interface stays in Indonesian, so on-screen text is shown as it appears to users.

REVISION 1

Manage sizes (master data)

Sizes were a fixed S, M, L, XL list in the code. The product form now reads them from the master, so the XXL added below appears immediately.

Before

Ukuran

Tambah varian ukuran (S/M/L/XL) dengan stok masing-masing

SIZE	LINGKAR DADA	PANJANG	LENGAN	STOK
S	cm	cm	cm	0
M	cm	cm	cm	0
L	cm	cm	cm	0
XL	cm	cm	cm	0

After

Ukuran

Tambahkan varian ukuran beserta stok masing-masing.

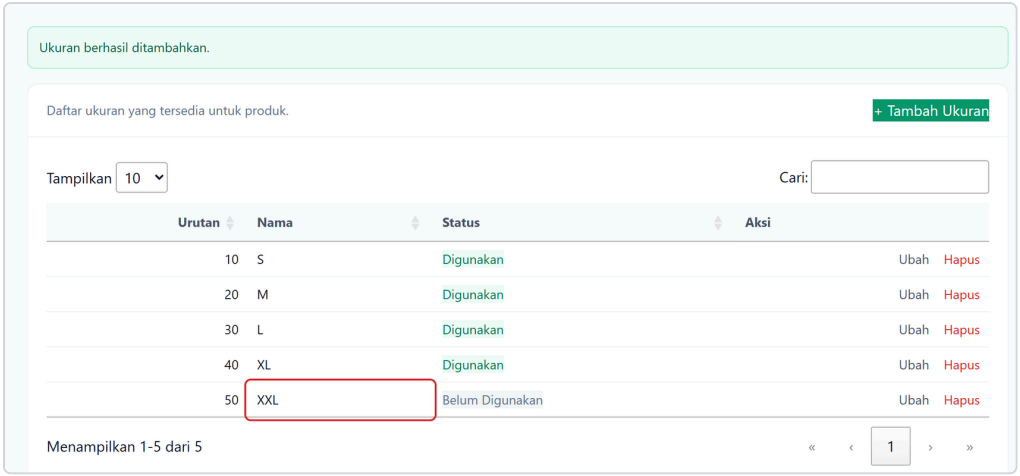
UKURAN	LINGKAR DADA	PANJANG	LENGAN	STOK
S	cm	cm	cm	0
M	cm	cm	cm	0
L	cm	cm	cm	0
XL	cm	cm	cm	0
XXL	cm	cm	cm	0

REVISION 1

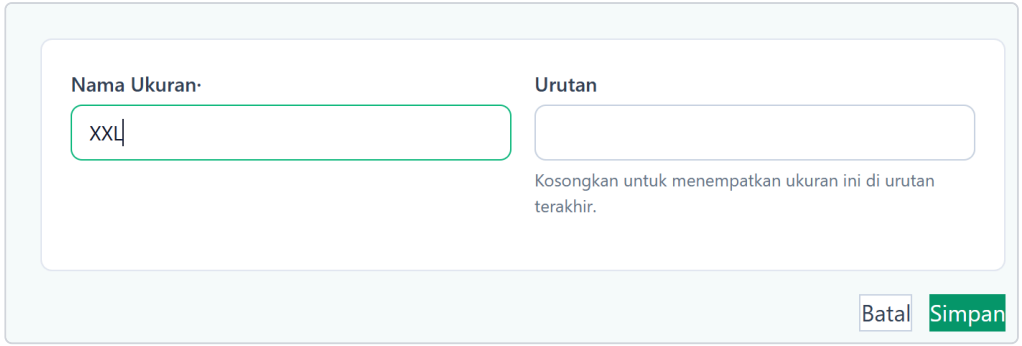
New Ukuran page

Add, edit and delete sizes. The Status column shows whether a size is in use; a size in use cannot be renamed or deleted.

After: size list, with XXL just added



After: add size form

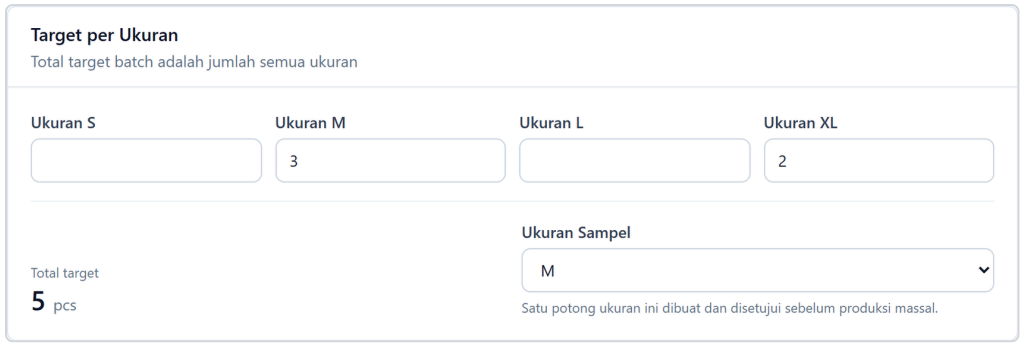


REVISION 2

Pick sizes when creating a batch

Before, every size of the product had an input. Now the user ticks only the sizes to produce, and the total and sample size follow the ticks. Here only M and XL are produced.

Before



After



REVISION 3

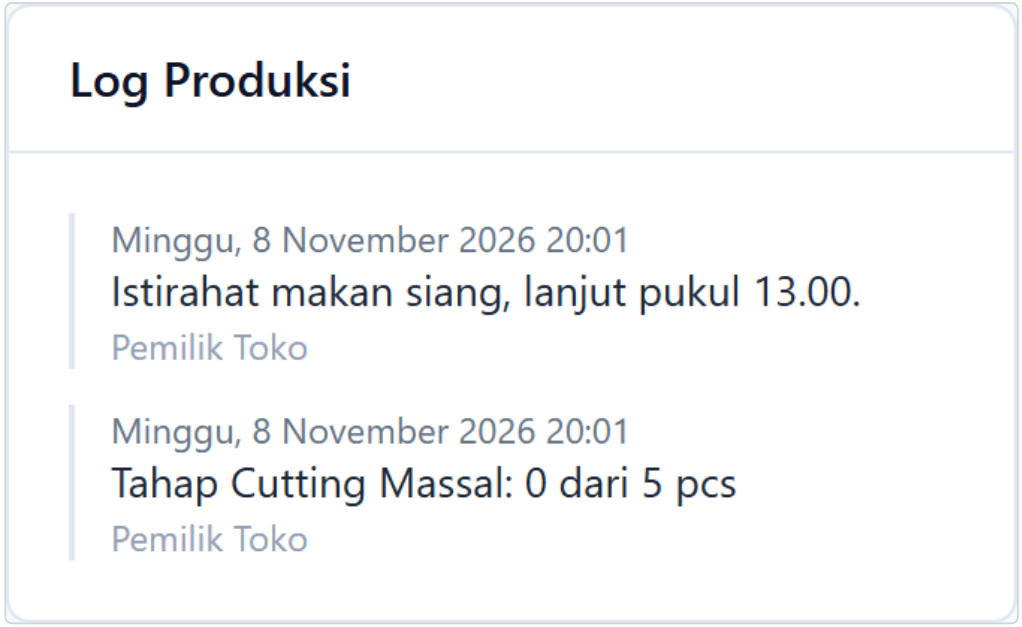
Daily production log with Pause

New on the batch detail page, so there is no before image. Each Pause adds an entry in WIB. The first entry has a typed note; the second was left blank, so the current stage was recorded automatically. The batch status does not change.

After: Pause card



After: Log Produksi, oldest first

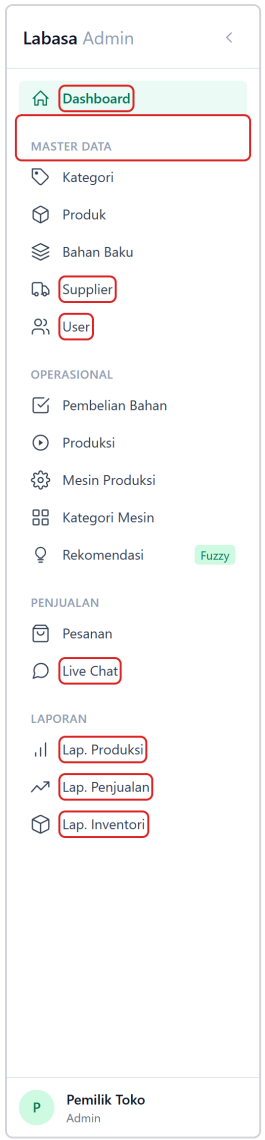


REVISION 4

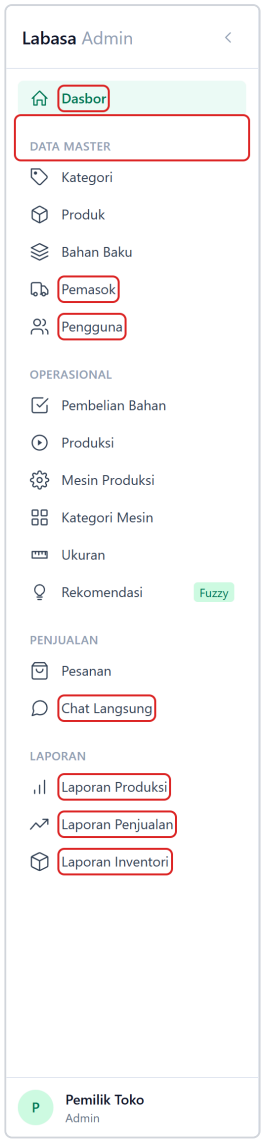
Wording: admin menu

Dashboard → Dasbor, Master Data → Data Master, Supplier → Pemasok, User → Pengguna, Live Chat → Chat Langsung, and report labels written in full.

Before



After

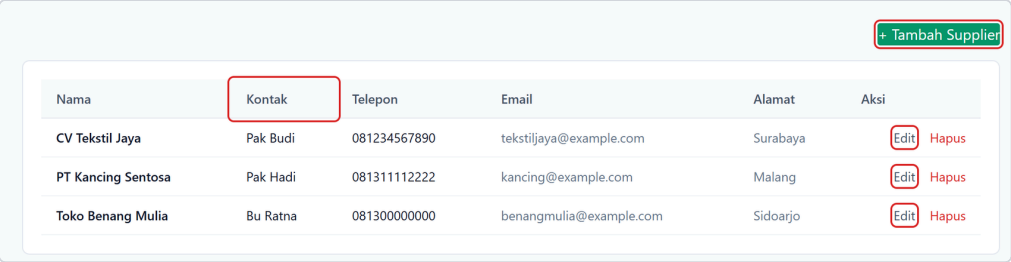


REVISION 4

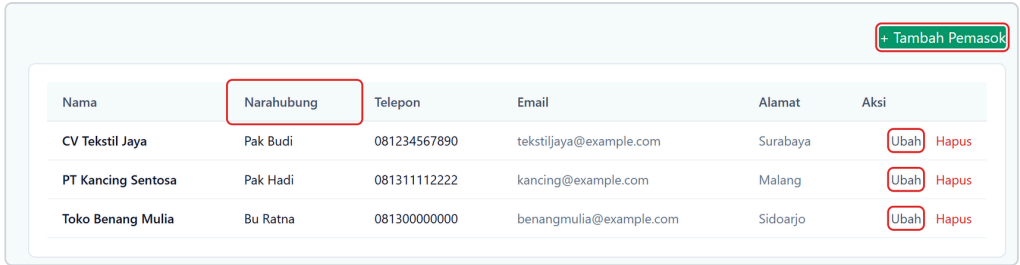
Wording: supplier list

Supplier → Pemasok, Kontak → Narahubung, Edit → Ubah.

Before



After

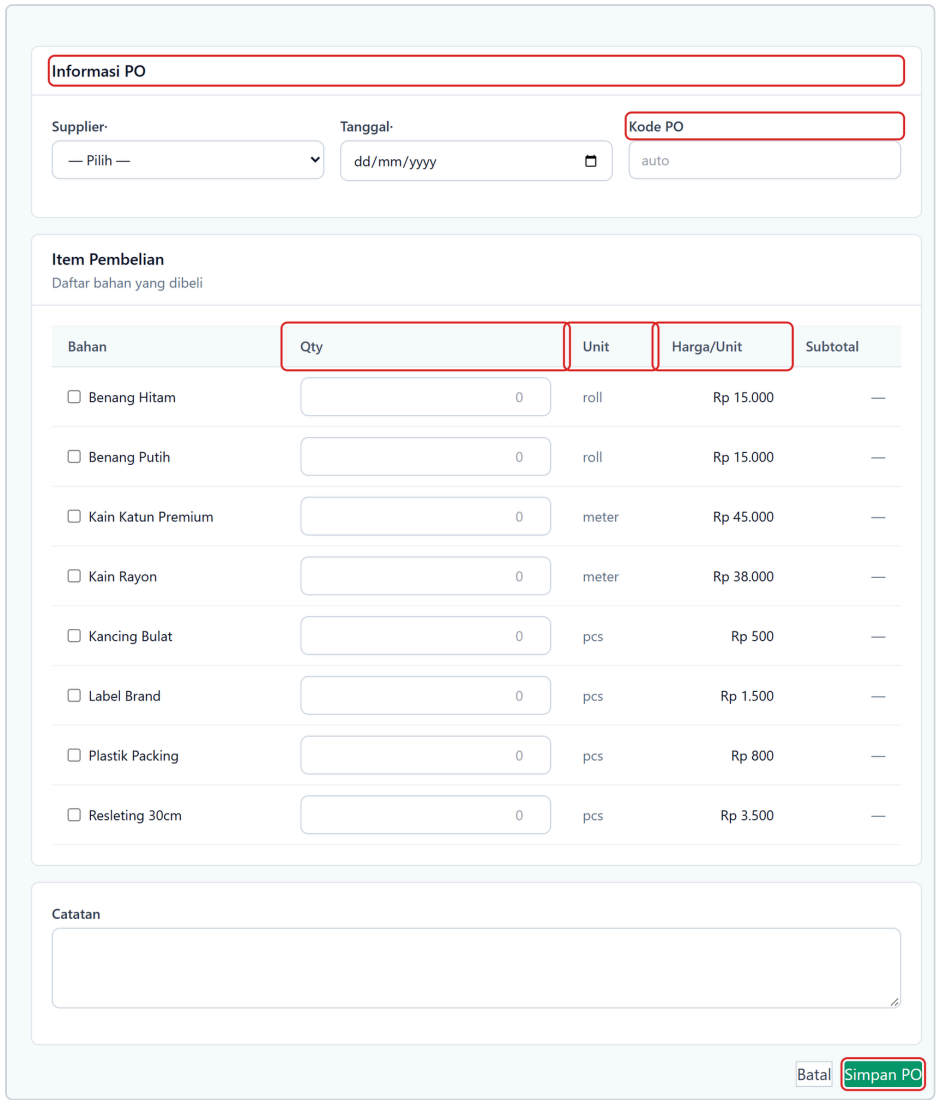


REVISION 4

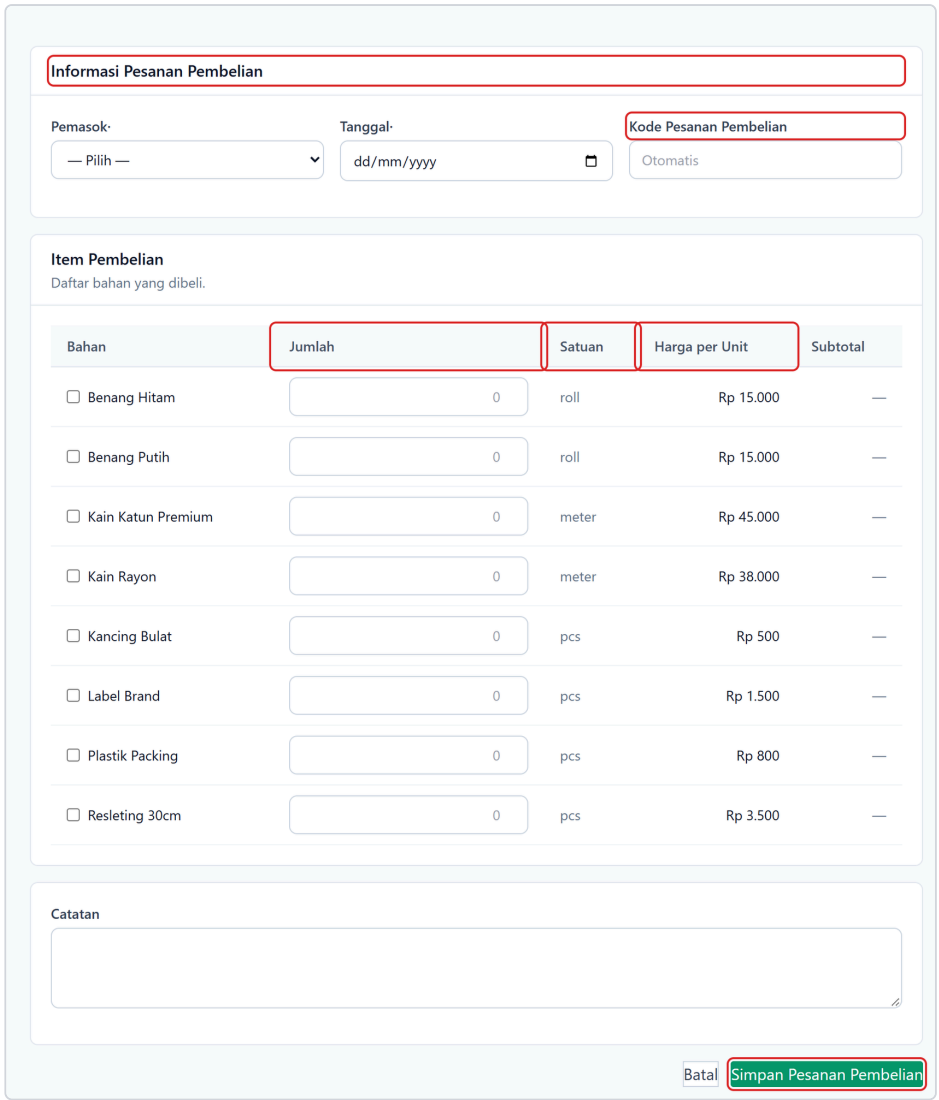
Wording: purchase order form

PO → Pesanan Pembelian, Supplier → Pemasok, Qty → Jumlah, Unit → Satuan, Harga/Unit → Harga per Unit.

Before



After

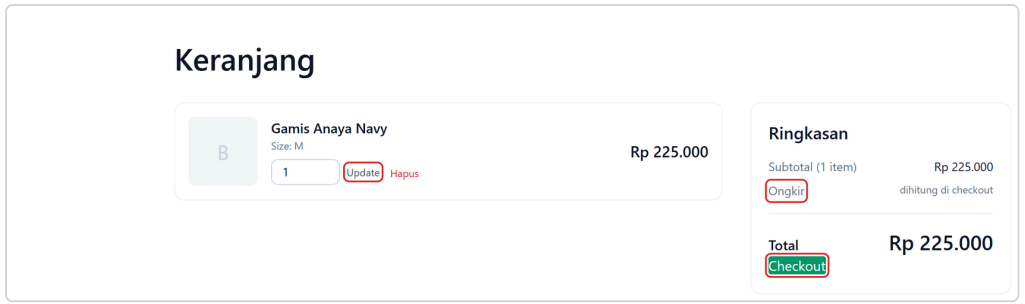


REVISION 4

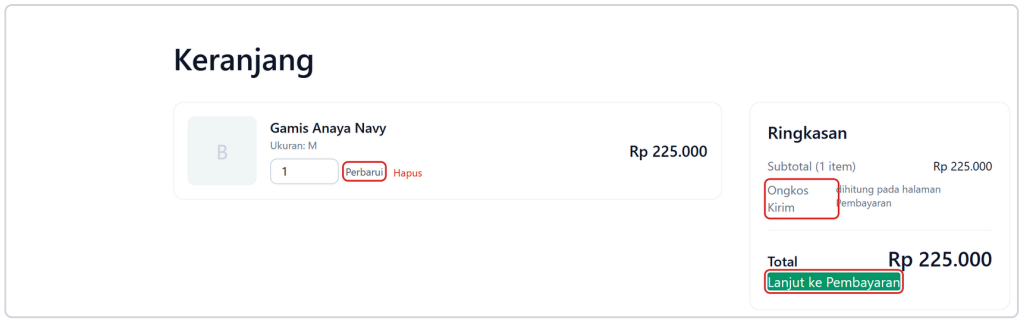
Wording: customer cart

Update → Perbarui, Ongkir → Ongkos Kirim, Size → Ukuran, Checkout → Lanjut ke Pembayaran.

Before



After

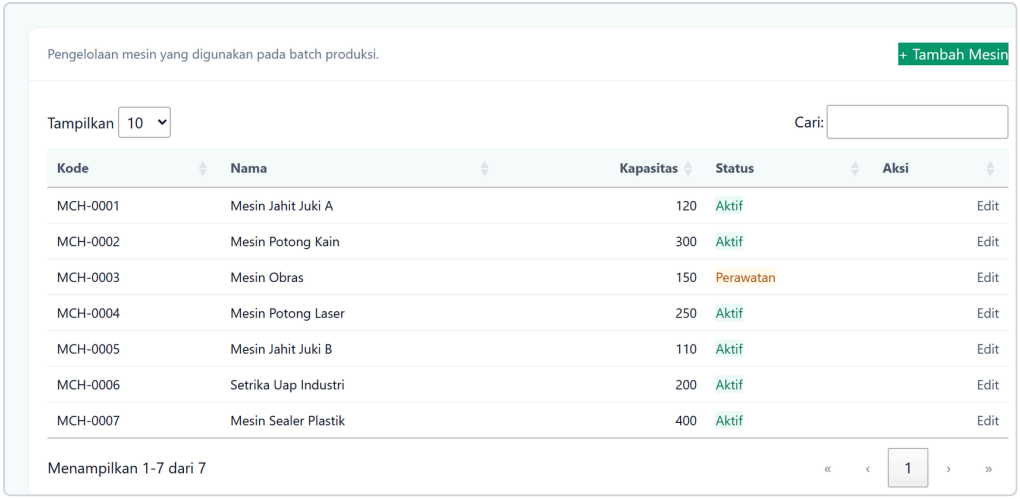


REVISION 5

Machine ID: machine list

The Kode column (MCH-0001) becomes ID Mesin (MSN-001), generated by the system and sorted by default.

Before



Pengelolaan mesin yang digunakan pada batch produksi.

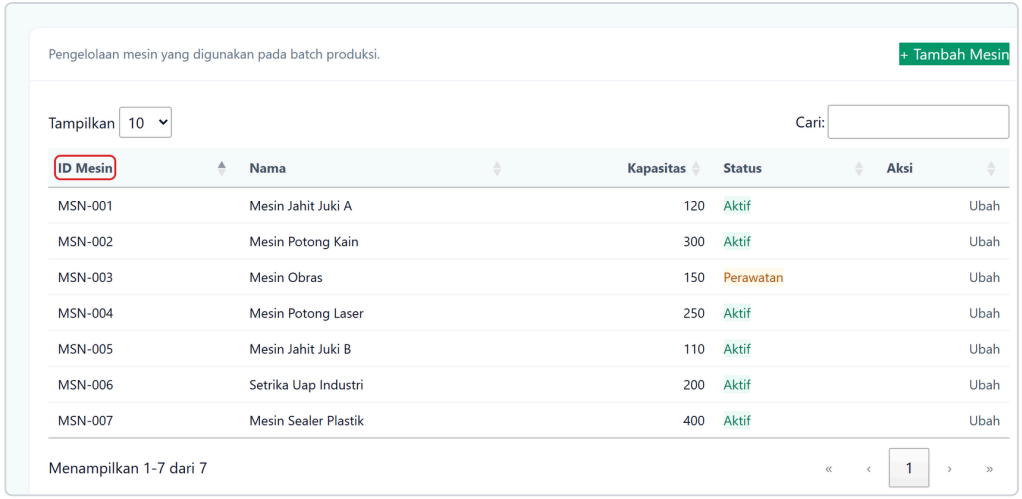
+ Tambah Mesin

Tampilkan 10 Cari:

Kode	Nama	Kapasitas	Status	Aksi
MCH-0001	Mesin Jahit Juki A	120	Aktif	Edit
MCH-0002	Mesin Potong Kain	300	Aktif	Edit
MCH-0003	Mesin Obras	150	Perawatan	Edit
MCH-0004	Mesin Potong Laser	250	Aktif	Edit
MCH-0005	Mesin Jahit Juki B	110	Aktif	Edit
MCH-0006	Setrika Uap Industri	200	Aktif	Edit
MCH-0007	Mesin Sealer Plastik	400	Aktif	Edit

Menampilkan 1-7 dari 7 « ‹ 1 › »

After



Pengelolaan mesin yang digunakan pada batch produksi.

+ Tambah Mesin

Tampilkan 10 Cari:

ID Mesin	Nama	Kapasitas	Status	Aksi
MSN-001	Mesin Jahit Juki A	120	Aktif	Ubah
MSN-002	Mesin Potong Kain	300	Aktif	Ubah
MSN-003	Mesin Obras	150	Perawatan	Ubah
MSN-004	Mesin Potong Laser	250	Aktif	Ubah
MSN-005	Mesin Jahit Juki B	110	Aktif	Ubah
MSN-006	Setrika Uap Industri	200	Aktif	Ubah
MSN-007	Mesin Sealer Plastik	400	Aktif	Ubah

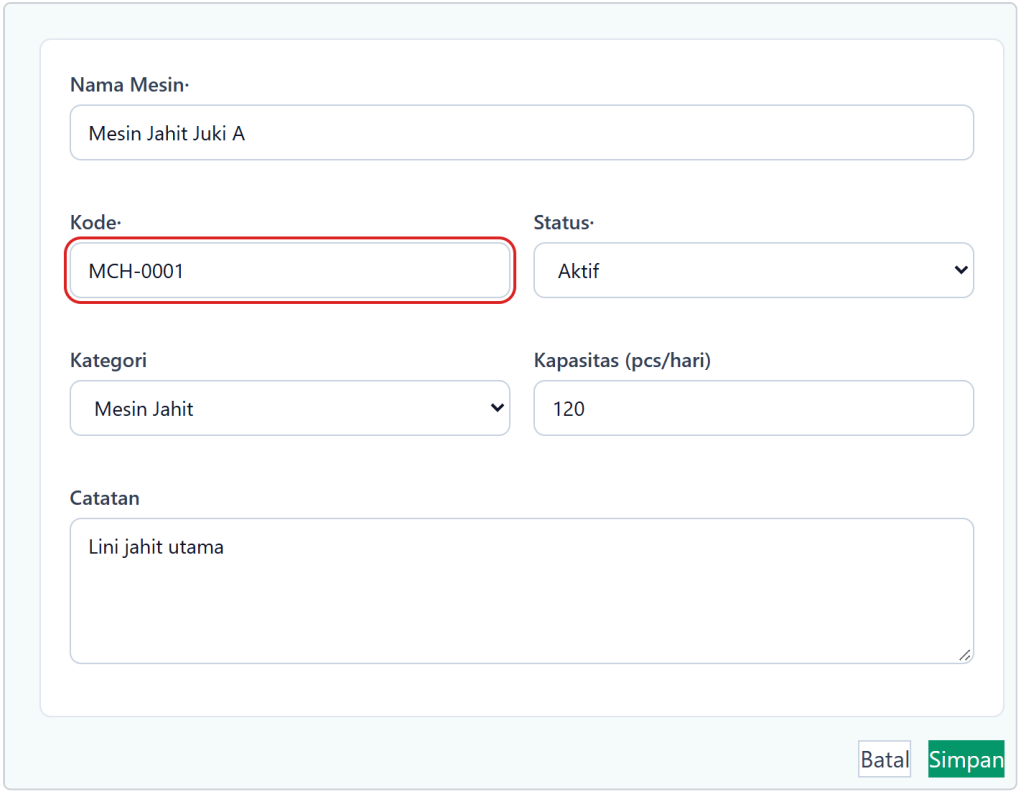
Menampilkan 1-7 dari 7 « ‹ 1 › »

REVISION 5

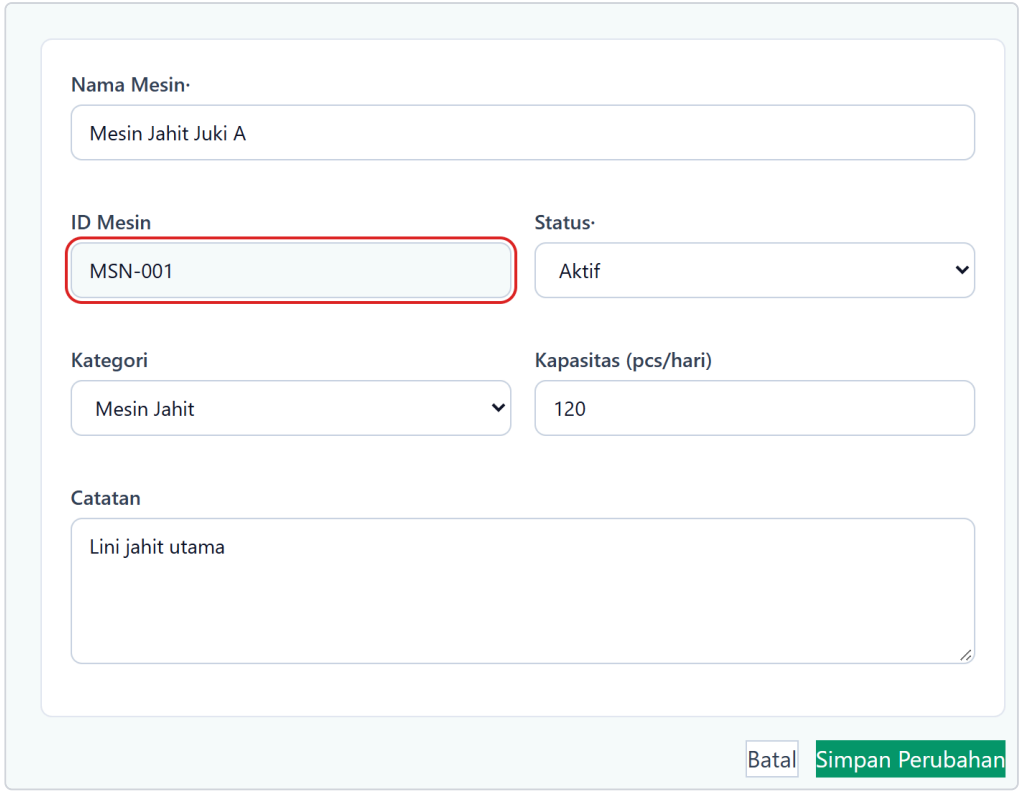
Machine ID: edit form

The code was a free text field. It is now read-only, so it cannot be changed by mistake.

Before



After

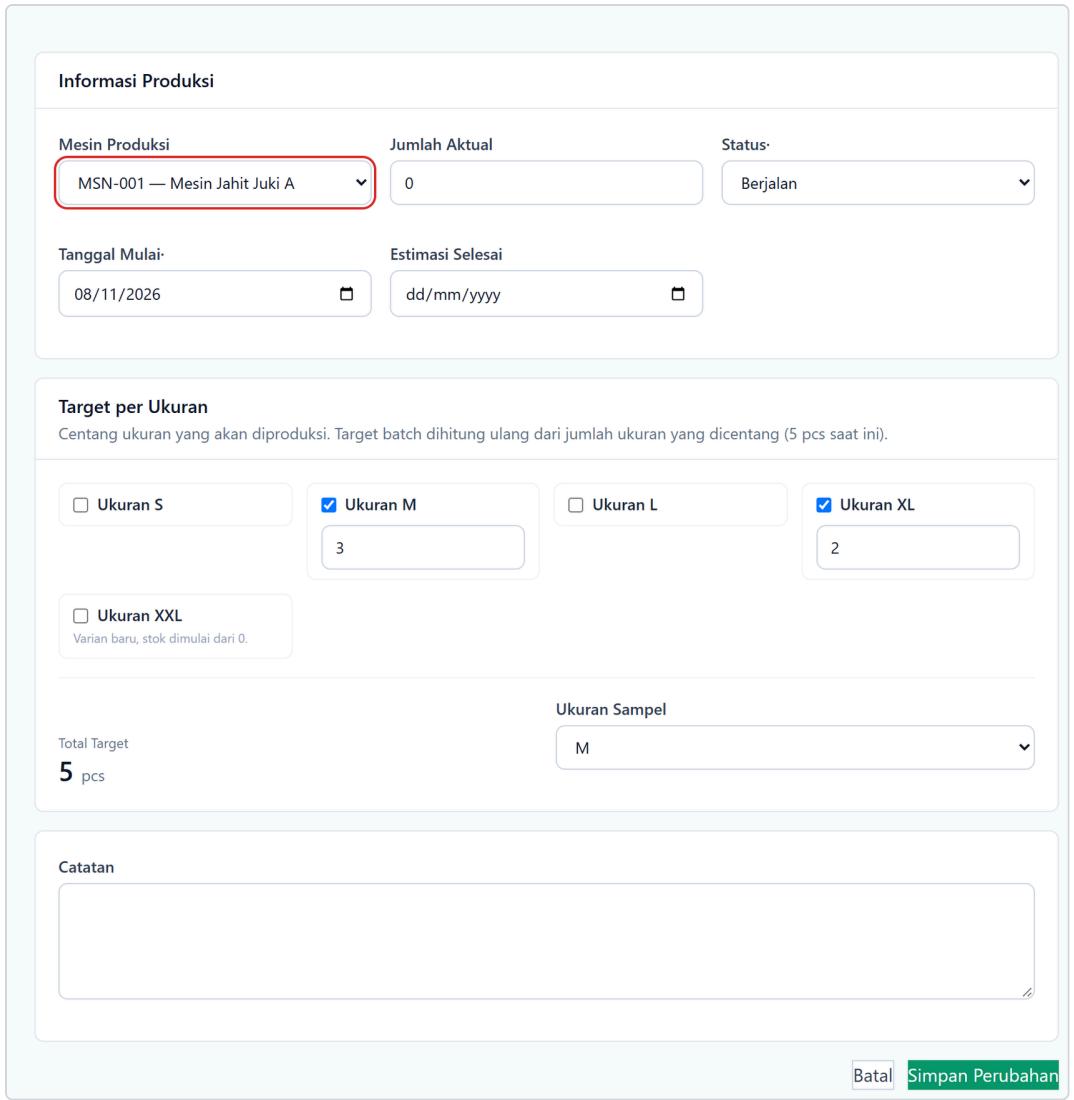


REVISION 5

Machine ID: selecting a machine

Machine selectors show the code with the name, for example on the batch edit form.

After: batch edit form

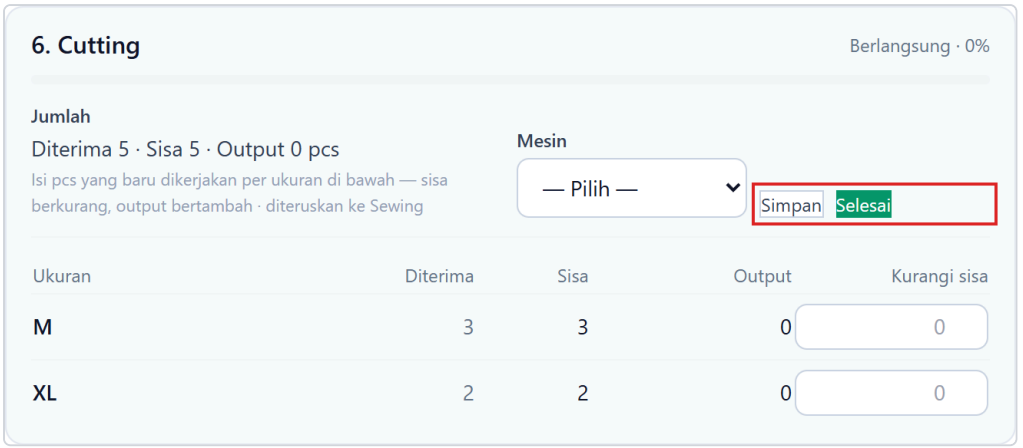


REVISION 6

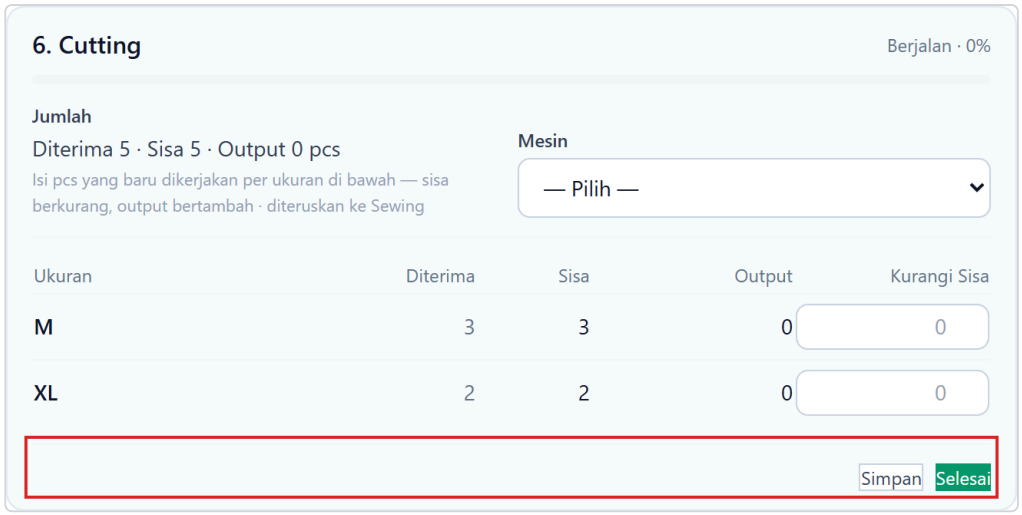
Buttons at the bottom: stage form

Simpan and Selesai sat beside the machine field, above the per-size table. They now close the form, so it reads top to bottom.

Before



After

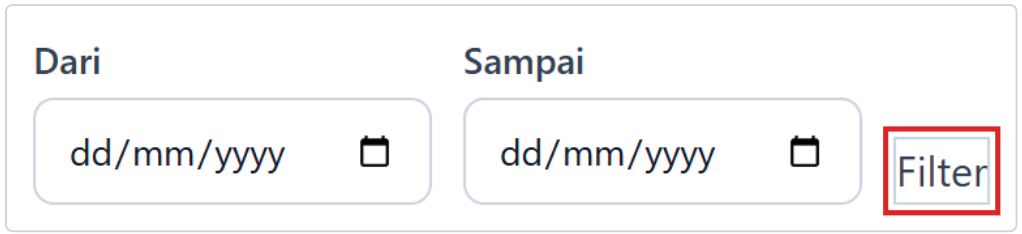


REVISION 6

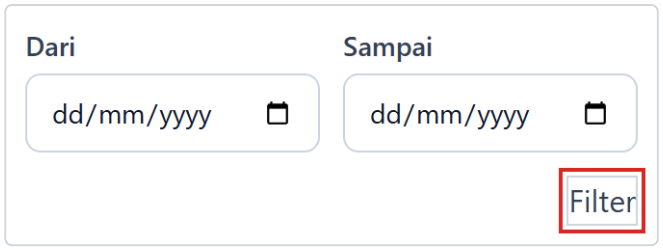
Buttons at the bottom: report filter and checkout

The same rule applied to the production report filter and the customer checkout.

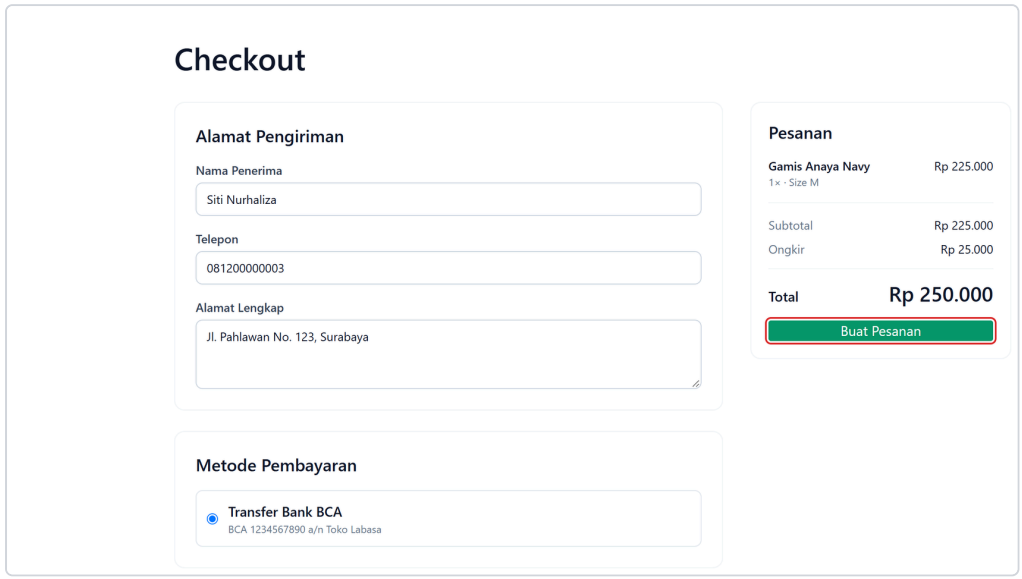
Before



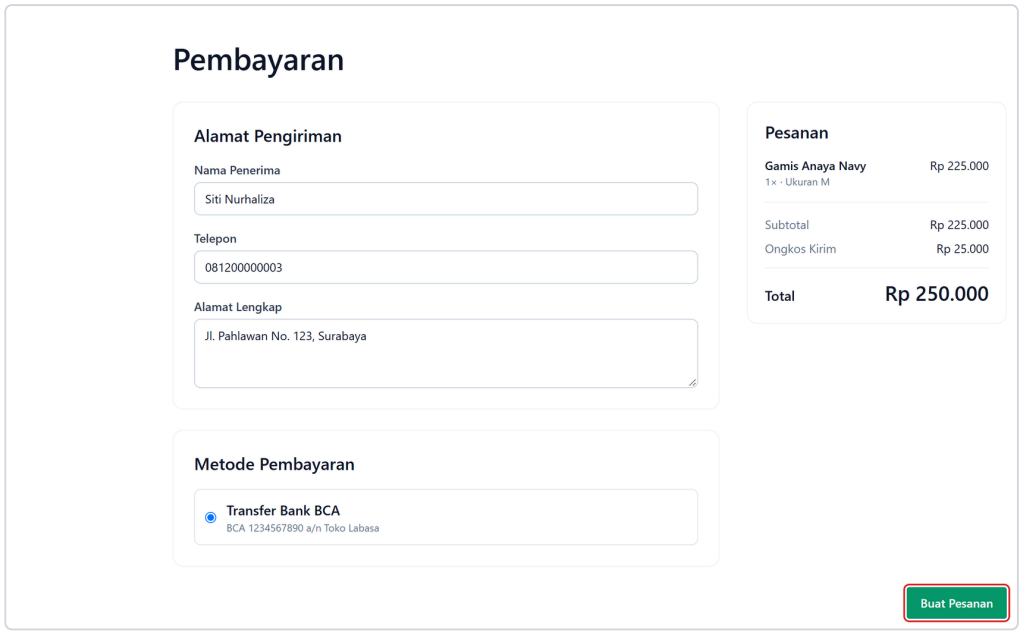
After



Before



After



REVISION 7

Fuzzy input follows product stock

Persediaan was a typed value. It now shows the selected product's actual stock as read-only, and the server ignores any value sent with the form.

Before

Input Parameter

Masukkan permintaan dan persediaan saat ini

Produk·

Gamis Anaya Navy

Permintaan (unit)·

5877

Permintaan untuk periode yang akan dihitung

Persediaan Saat Ini (unit)·

2370

Stok produk yang masih tersedia

Hitung Rekomendasi

After

Parameter Input

Masukkan permintaan; persediaan mengikuti stok produk.

Produk·

Gamis Anaya Navy

Permintaan (unit)·

5877

Permintaan untuk periode yang akan dihitung.

Persediaan Saat Ini (unit)

2,370

Diambil otomatis dari stok produk.

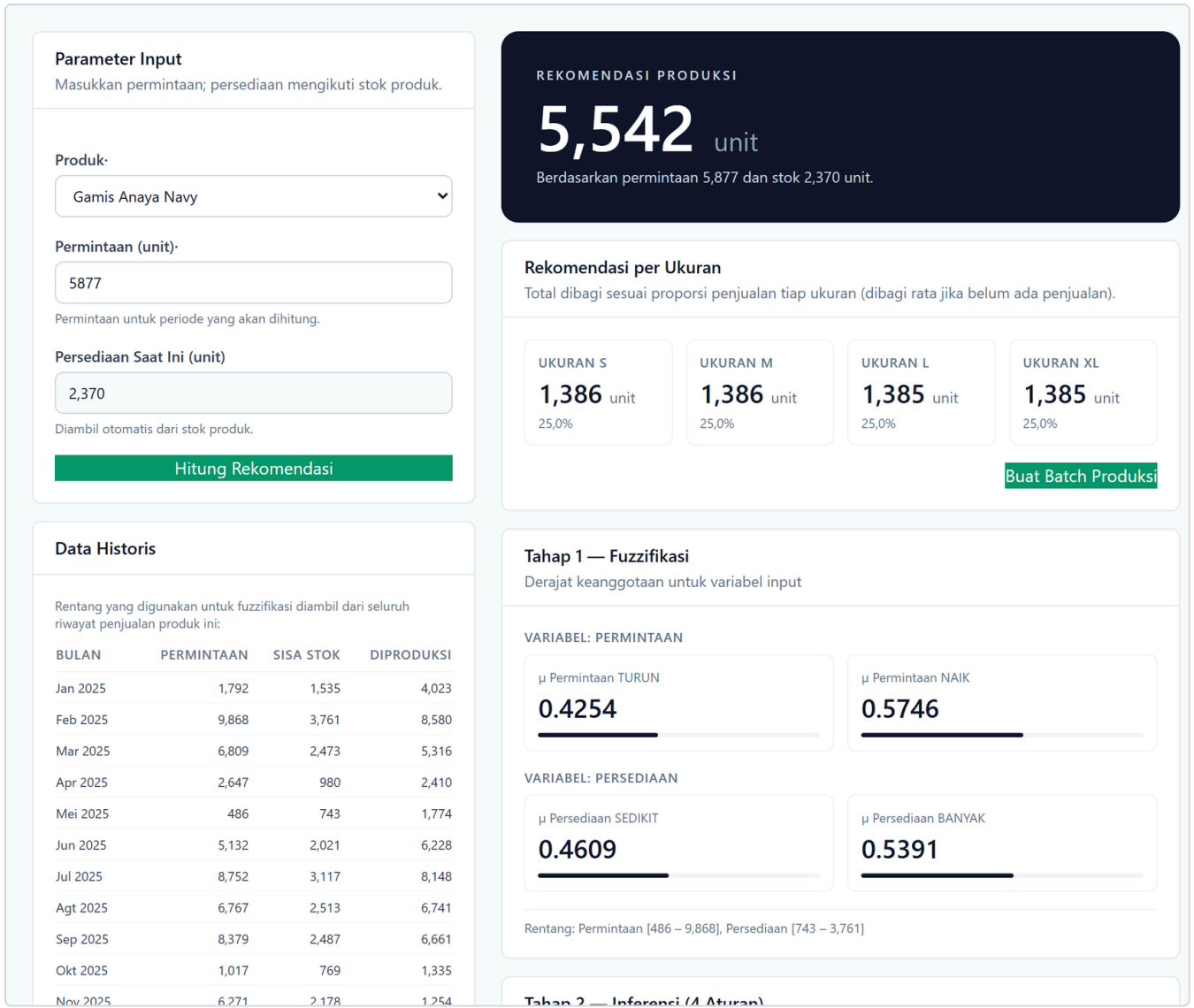
Hitung Rekomendasi

REVISION 7

Fuzzy result using product stock

With demand 5877 and product stock 2370, the worked example from the proposal gives a recommendation of 5,542 units. The image shows the top of the result page.

After: calculation result



Parameter Input

Masukkan permintaan; persediaan mengikuti stok produk.

Produk·

Gamis Anaya Navy

Permintaan (unit)·

5877

Permintaan untuk periode yang akan dihitung.

Persediaan Saat Ini (unit)

2,370

Diambil otomatis dari stok produk.

Hitung Rekomendasi

REKOMENDASI PRODUKSI

5,542 unit

Berdasarkan permintaan 5,877 dan stok 2,370 unit.

Rekomendasi per Ukuran

Total dibagi sesuai proporsi penjualan tiap ukuran (dibagi rata jika belum ada penjualan).

UKURAN S	UKURAN M	UKURAN L	UKURAN XL
1,386 unit	1,386 unit	1,385 unit	1,385 unit
25,0%	25,0%	25,0%	25,0%

Buat Batch Produksi

Data Historis

Rentang yang digunakan untuk fuzzifikasi diambil dari seluruh riwayat penjualan produk ini:

BULAN	PERMINTAAN	SISA STOK	DIPRODUKSI
Jan 2025	1,792	1,535	4,023
Feb 2025	9,868	3,761	8,580
Mar 2025	6,809	2,473	5,316
Apr 2025	2,647	980	2,410
Mei 2025	486	743	1,774
Jun 2025	5,132	2,021	6,228
Jul 2025	8,752	3,117	8,148
Agt 2025	6,767	2,513	6,741
Sep 2025	8,379	2,487	6,661
Okt 2025	1,017	769	1,335
Nov 2025	6,271	2,178	1,254

Tahap 1 — Fuzzifikasi

Derajat keanggotaan untuk variabel input

VARIABEL: PERMINTAAN

μ Permintaan TURUN	μ Permintaan NAIK
0.4254	0.5746

VARIABEL: PERSEDIAAN

μ Persediaan SEDIKIT	μ Persediaan BANYAK
0.4609	0.5391

Rentang: Permintaan [486 – 9,868], Persediaan [743 – 3,761]

Tahap 2 — Inferensi (4 Aturan)

Issues seen during this check

These were found while taking the screenshots and are **not fixed**.

Issue	Where	Detail
Buttons without padding	Admin forms and lists; some customer pages	Visible in the images above, for example the cramped Simpan and + Tambah buttons. Views use `btn-primary` without `btn`, and the padding is on `.btn`. It appears in both Before and After, so these revisions did not cause it.
Thousands separator	Recommendation page	Numbers show as 2,370 and 5,542. Indonesian format would be 2.370 and 5.542.
Sample size on edit	Batch edit form	Unticking the current sample size while its sample is in progress can change the sample size.
"6 tahap" claim	Landing page	The text says 6 quality-check stages, while the system has 5 production stages.